1. **Human Reservoir**

   a. Cases - There are a number of important pathogens that are specifically adapted to man, such as: measles, smallpox, typhoid, meningococcal meningitis, gonorrhea and syphilis. The cycle of transmission is from human to human.

   b. Carriers - is a person who has become infected with a pathogen, but does not show any signs or symptoms.

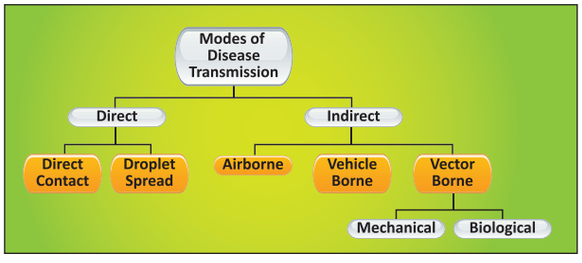
2. **Animal Reservoir** - the source of infection may sometimes be animals and birds.

3. **Nonliving things** - soil and inanimate matter can also act as reservoir of infection. eg. Tetanus in soil.

11.3 Mode of Transmission

Infectious diseases are transmitted from person to person by direct or indirect contact. Certain types of viruses, bacteria, parasites, and fungi can cause infectious disease. Malaria, measles, and respiratory illnesses are examples of infectious diseases.

Susceptible Host - A person who lacks resistance to a particular pathogenic agent to prevent disease if or when exposed.



I. Direct Transmission

1. Direct Contact - Infection may be transmitted directly from skin to skin, mucosa to mucosa, mucosa to skin of others or same person. Example: skin-to-skin contact as by touching, kissing or sexual intercourse. Diseases transmitted - STD and AIDS, leprosy, leptospirosis, skin and eye infections.

2. Droplet infection -This is direct projection of a spray of droplets of salaiva and nasopharyngeal secretions (airborne droplets of saliva or sputum) containing infectious organisms. The spray of droplets during coughing and sneezing can spread an infectious disease.

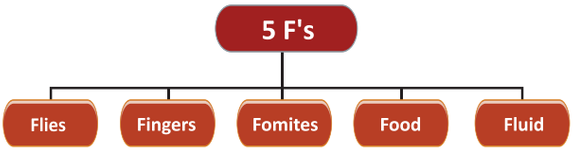
3. Contact with soil - The disease agent may be acquired by direct exposure of susceptible tissue to the disease agent in soil, compost or decaying vegetable matter. Examples: hookworm larvae, tetanus, mycosis etc.

4. Inoculation into skin or mucosa - Rabies virus by dog bite, Hep. B virus through contaminated needles and syringes etc.

5. Transplacental (or vertical) transmission - TORCH agents - (Toxoplasma Gondii, Rubella virus, Cytomegalo virus, and Herpes virus). Varicella virus, Syphilis, Hepatitis B, Coxsackie B and AIDS. Some of the non-living agents (e.g., thalidomide, diethylstilbestrol) can also be transmitted vertically (mother to child) which will affect the embryo and causes malformations in the foetus.

11.4 Indirect Transmission

This embraces a variety of mechanisms including the traditional 5 F's, such as



1. Vehicle-borne - An indirect transmission of an infectious agent that occurs when a vehicle. (or formites) touches a persons body or is ingested

2. Vector - borne - Vector is defined as an arthropod or any living carrier that transports an infectious agent to a susceptible individual. Infectious agents are transmitted

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